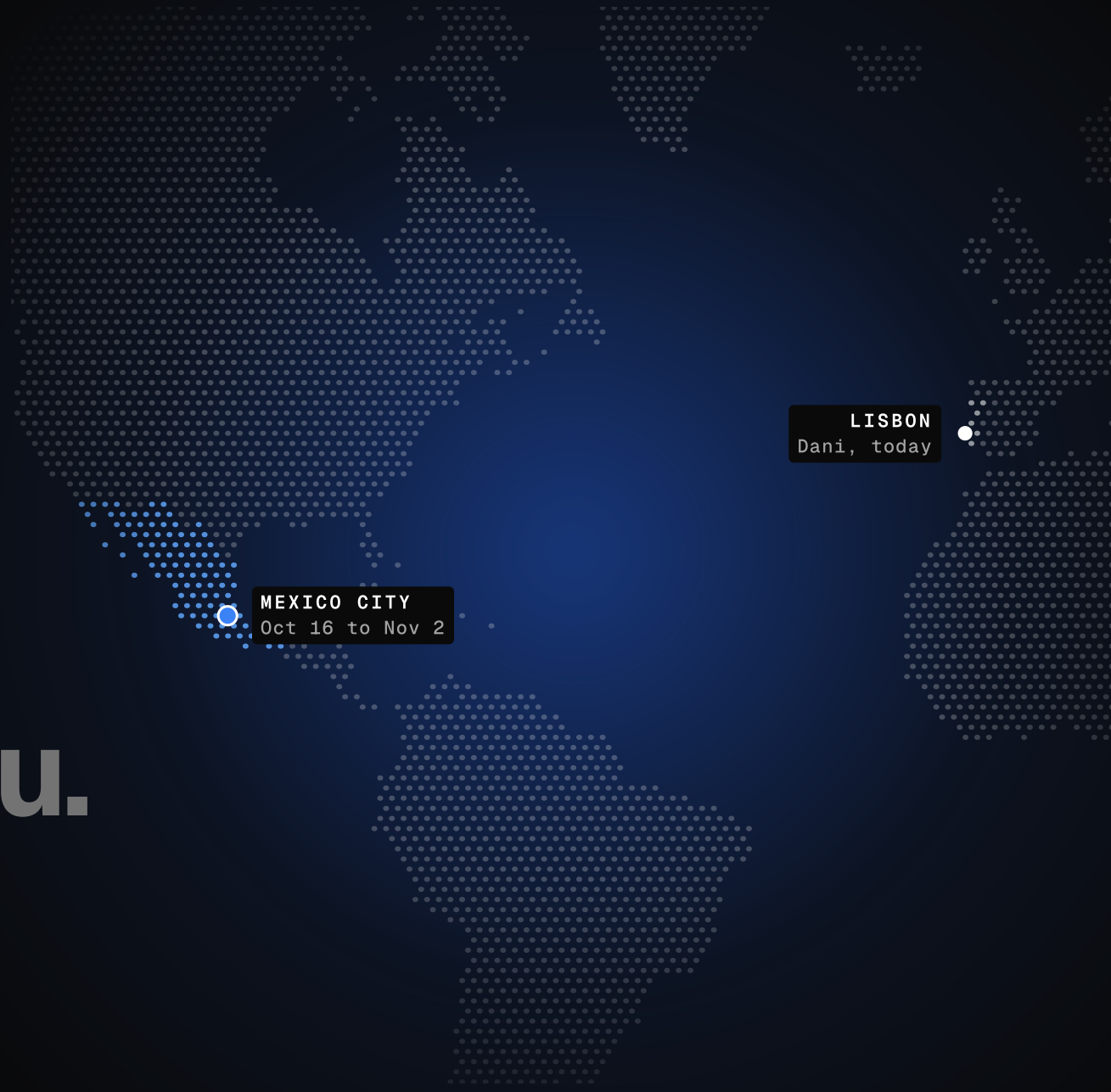
NOMAD.OS

ASSIGNMENT 5 · AI INTERACTION PROTOTYPE

Books for you.
Never without you.

A travel agent inside NOMAD.OS. It books trains, hotels, and dinner, and asks before it pays or shares anything.



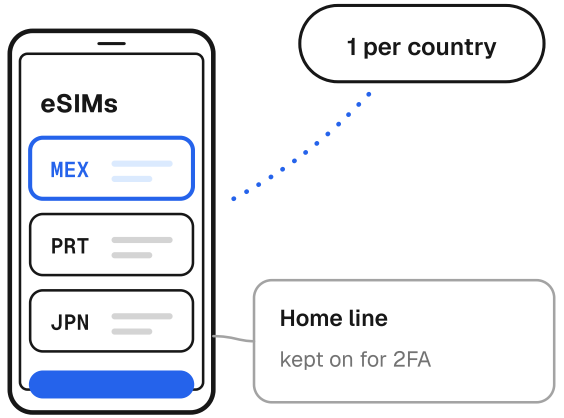
IDSN 523 · Mitch Kostelecky and Ian Harris · Fall 2026

EXISTING PRODUCT AND SELECTED EXPERIENCE

Airalo.
An eSIM for every country.

The closest real product to our Numbers layer: a data plan you download for each country, live in minutes.

30M+ travelers · 200+ destinations · local, regional, and global plans



AIRALO'S FLOW TODAY

01 **Pick a destination**
Country or region

02 **Buy a data plan**
Priced by data and days

03 **Install the eSIM**
Before you fly

04 **Connect on arrival**
Turn on the eSIM line

05 **Top up or renew**
Add data any time

SELECTED EXPERIENCE

The three days before a new country: getting connected, booked, and set up to work.

WORKS WELL, KEEP IT

- ✓ Fully digital, live in minutes, no SIM swap
- ✓ Home line stays on for calls and codes

WHERE IT STOPS

- × Ends at data. Stays, trains, and dinner happen in other apps
- × Line, roaming, and code tradeoffs are left to you

Dani lands in three days.
Every step is a different app.

Dani
Remote worker, moves every few weeks

GOAL

Land in Mexico City on Oct 16 connected, booked, and ready to work.

DECISIONS

Which listing is real, which card pays, what to share, which line gets bank codes

NEEDS TO KNOW

Honest reviews, tested Wi-Fi, refund terms, which banks flag VPN logins

Illustrative composite from our A4 study of 27 public posts

CURRENT FLOW, TUE OCT 13

Buy an eSIM Airalo	Works well: live in minutes, home line kept on.	Keep	
Pick a line for bank codes Help articles	Roaming or codes? Advice conflicts: three answers in one week.	E14	E23
Book 17 nights Booking site	Is the listing real? Is the Wi-Fi good enough to work from?		
Book a train, dinner, delivery Three more apps	Card and passport typed in again. No record of who has what.		
Land and log in Bank app	A VPN login can freeze the bank, or a work account.	E13	E26

Worth keeping: Airalo's speed, real reviews, free cancellation, and Dani's own choices.

OPPORTUNITY STATEMENT

When **a remote worker** is trying to **land in a new country ready to work**, they experience **a pile of bookings and setup decisions spread across separate apps** because **each app sells one piece and leaves the risks in between to guesswork**. There is an opportunity to help them **book what they need and arrive set up, approving every payment and every detail shared**.

AI IS NOT AUTOMATICALLY THE ANSWER. WE COMPARED THREE OPTIONS.

A better checklist

Same list for everyone. It can't know that Wise flags VPN logins in Mexico, or that a listing wants a wire.

Kept as the fallback: every fix shows how to do it yourself.

A better booking site

Books one thing well, but can't see the card, the network, or what other apps already have.

Simpler, but the risks in between stay with the traveler.

An AI agent

Our pick

Reads a plain request with dates, card, and network together, then prepares each booking to approve.

Worth it only with a confirm step and hard limits.

Primary role: agent.
Secondary role: interpreter.

AGENT

Books trains, hotels, tables, and delivery from one chat.

Takes actions for the user within defined boundaries.

Agent tab, screen S8a

INTERPRETER

Check my setup explains each fix for the next country.

Makes complex information understandable.

Check my setup, screen S8b

Why it fits

Getting ready is many small actions across services. An agent can take them, inside limits Dani sets.

What it needs

The request, trip dates, the card it pays with, and network status. Banks, lines, and sign-ins only if Dani shares them.

What it produces

Shortlists with tested Wi-Fi and reviews, risk flags with reasons, confirm cards, and bookings in Trips.

Value beyond Airalo

One place to book and arrive set up. Scams flagged before anyone pays. A record of what each booking shared.

Where it enters

The Agent tab, any time, and Check my setup after a profile downloads. Nothing runs in the background.

Stays with Dani

Choosing, paying, what gets shared, cancelling, and anything inside a bank.

HUMAN AND AI RESPONSIBILITY MAP

The agent prepares. Dani decides.

STAGE	DANI	AGENT	SHARED DECISION
Initiation	Opens the Agent tab and asks, or taps a tile	Greets with trip context. Never starts on its own	After a hotel booking, suggests a setup check. Dani chooses to open it
Input and interpretation	Says what's needed. Picks what the setup check can see	Matches the service, fills in city, dates, and card	Missing details, like a time or party size, get asked, not guessed
Generation or action	Compares options and reads reviews	Shortlists options. Leaves out risky ones and says why	Dani picks one. Change brings the options back
Decision or approval	Confirms, changes, or cancels. Face ID over $100	Shows cost, card, terms, and what's shared first	The confirm card: the agent fills it in, Dani approves it
Completion and review	Checks Trips. Can cancel any time	Books, files it in Trips, says how it's protected	Setup fixes: the agent suggests, Dani approves, skips, or undoes

HUMAN JUDGMENT

Which option, paying, what's shared, cancelling

AGENT PREPARES

Shortlists, risk flags, confirm cards, the setup checklist

AGENT DOES ALONE

Searching, filtering, filing bookings in Trips

EXPLICIT PERMISSION

Anything that costs money or shares details. Face ID over $100

EXPLAINS ITSELF

When it leaves something out, says no, or suggests a fix

DANI STEPS IN

Change or Cancel on every card. Cancel later from Trips

IF IT'S WRONG

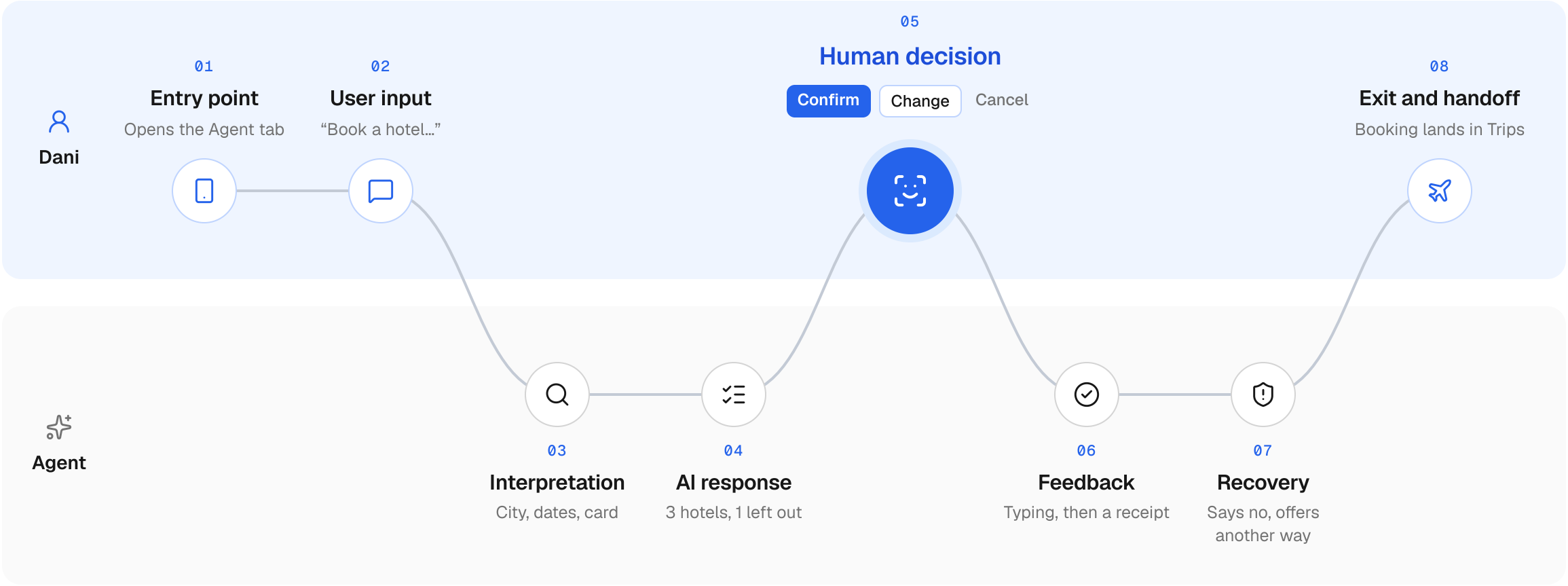
Dani approved the card, so NOMAD.OS answers for what the card showed

NEVER ALLOWED

Pay by bank transfer or off the booking site. Change bank settings. Hide or fake location. Act without an OK.

PROPOSED AI INTERACTION FLOW

Dani starts it and decides it.
The agent does the work, then hands it back.



PROTOTYPE: BOOKING PATH

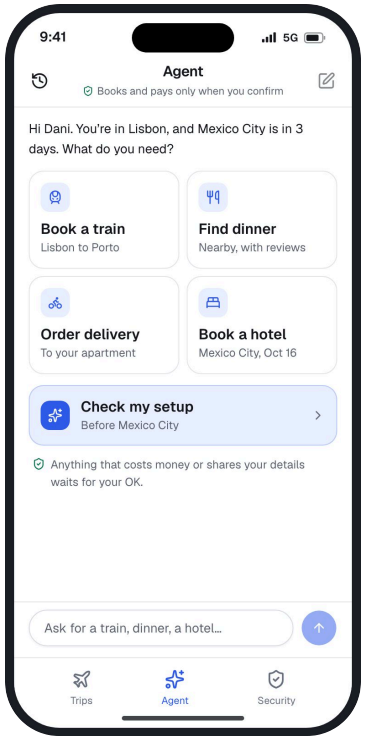
Ask, compare, approve.
Nothing books until Dani confirms.

20 Every booking ends in a confirm card. Nothing books on its own.

21 Says exactly what gets shared, and with whom.

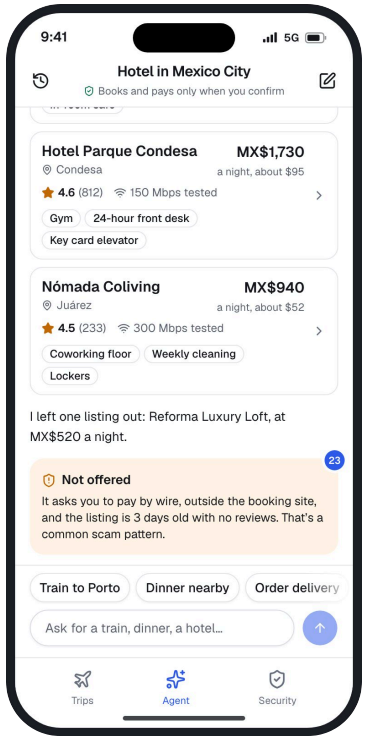
22 Big purchases need Face ID, not just a tap.

23 Won't pay by transfer or off the booking site. Says why.



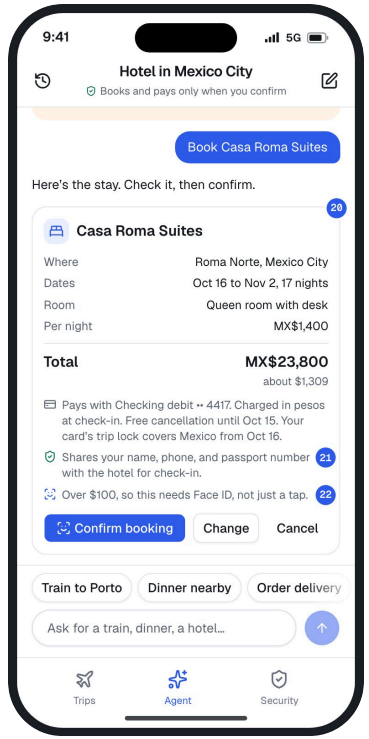
01 Ask

Tiles or plain words. The header says it books and pays only when Dani confirms.



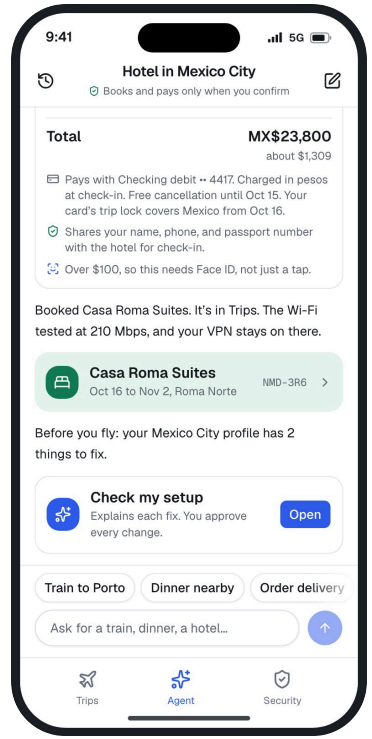
02 Compare

Wi-Fi tested by other nomads. A listing that wants a wire is left out, with why.



03 Approve

Dates, total in pesos and dollars, the card, and what the hotel gets.



04 Booked

Filed in Trips with a reference. Then one next step: 2 fixes before the flight.

Clickable prototype with scripted agent replies. Screen S8a, at **#agent**

Control stays with Dani.
Even when the agent says no.

STATES WE BUILT

Booking

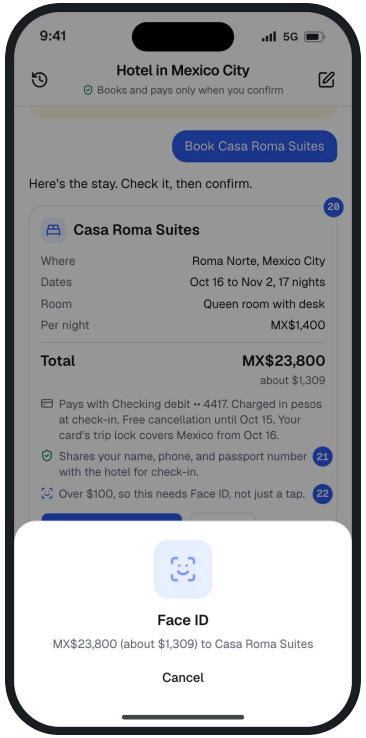
Options · Typing · Confirm · Face ID · Booked · Changed · Cancelled

Safety

Not offered · Refused · Can't help

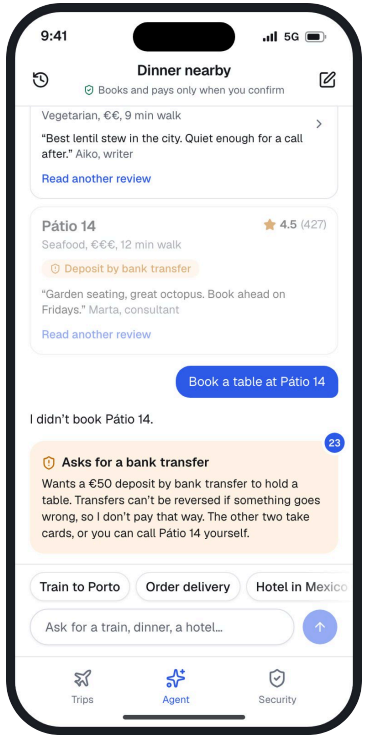
Setup

Share · Checking · Approve · Skip · Undo



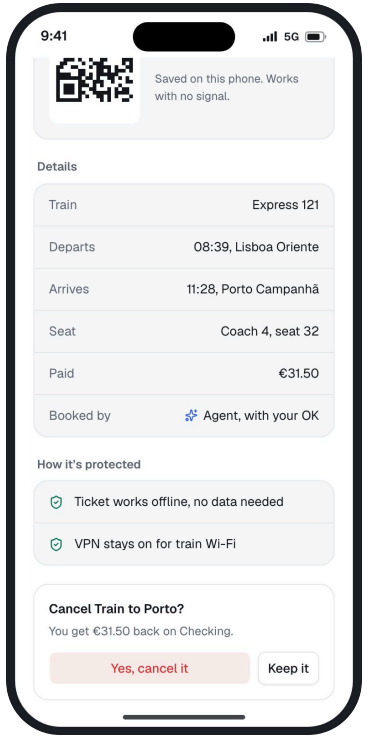
01 Verify

Over $100, so Face ID with the amount and the hotel. Cancel backs out.



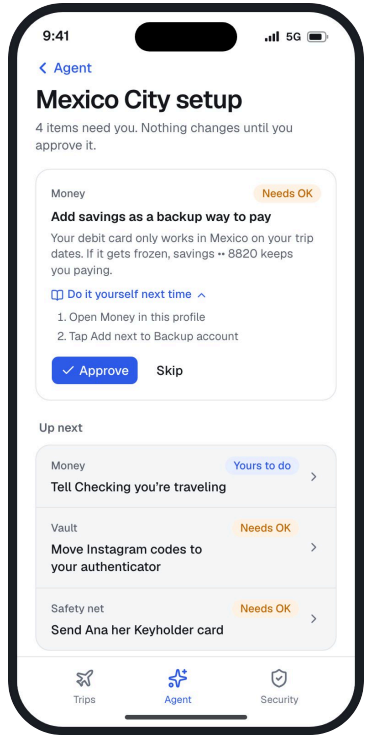
02 Refuse

A €50 deposit by transfer. It won't pay that way, says why, and offers two others.



03 Undo

Any booking cancels from Trips, with the refund and card spelled out.



04 Learn

Each fix says why and how to do it yourself. Approve, skip, or undo.

HUMAN FLOURISHING

Agency and learning.
Help that doesn’t take over.

	OPPORTUNITY	RISK	SAFEGUARD
Agency and autonomy Does it keep meaningful choice and control?	Dani makes every choice that matters: which option, which card, what’s shared.	Confirm fatigue: tapping yes without reading. Or a shortlist that quietly narrows the choice.	Short confirm cards. Face ID only over $100, so it still means something. Anything left out is shown, with why.
Learning and capability Does it build knowledge, judgment, or skills?	Every flag and fix explains why, so Dani learns what a risky listing or setup looks like.	Dependence: Dani stops judging listings or setting up lines, and can’t travel without the agent.	Do it yourself next time, on every fix. Bank changes always stay with Dani.

THE TENSION

Filtering scams protects Dani, but every hidden listing is a choice made on Dani’s behalf. So nothing is hidden silently.

HOW WE’D TEST IT

Five nomads, with a teammate answering as the agent from the operator panel. Do people read the confirm card? Can they repeat a fix alone a week later?

APPENDIX A

Where the AI enters, and where it stops.

What activates it?

Dani opens the Agent tab, or taps Check my setup after a profile downloads. Nothing runs in the background.

Does it ask, or suggest?

Dani asks. Its one suggestion: after a hotel booking, it points to setup fixes, and Dani decides whether to open them.

What information does it use?

The request, trip city and dates, the card on file, and network status. The setup check sees only what Dani shares.

Does Dani know what it uses?

Yes. The share screen lists each source, and every confirm card says what gets shared, and with whom.

What does it produce?

Shortlists, risk flags with reasons, confirm cards, bookings in Trips, and a setup checklist.

What needs approval?

Anything that costs money or shares personal details. Over $100 also needs Face ID.

Can Dani edit, reject, or reverse it?

Change or Cancel on every card. Bookings cancel from Trips. Setup approvals can be undone, except a sent Keyholder card.

How does it show uncertainty?

Prices are about-amounts in dollars, Wi-Fi speeds come from other nomads' tests, and ratings show review counts.

What if it can't finish?

It says so, explains why, and offers another way: a different option, or the steps to do it yourself.

When does it stop?

After the receipt and one next step. It never chains bookings on its own.

How does control return?

The booking sits in Trips, the chat waits, and every thread is saved in history.

APPENDIX B · FROM ASSIGNMENT 4

Opportunity statements and value proposition.

AREA	OPPORTUNITY STATEMENT	HOW MIGHT WE
Network judgment	Members value **competence** and participate through **VPN and travel-router routines**, but experience tension between **fear-driven advice** and **unclear real risk**.	How might we help nomads judge network risk with confidence without reinforcing fear-driven tool buying?
Security that moves with you Selected	Members value **self-reliance and freedom of movement** and participate through **second phones, two numbers, and pre-departure prep**, but experience tension between **security systems that assume a home** and **lives spent crossing borders**.	How might we help nomads arrive in any country already set up, and recover from anywhere, without assuming a home or hiding where they are?
Money that keeps moving	Members value **freedom of movement** and participate through **multi-bank, multi-app money stacks**, but experience tension between **automated compliance** and **travel patterns that look suspicious**.	How might we help nomads keep access to money during a freeze without needing a home bank to fall back on?

VALUE PROPOSITION

For **digital nomads and remote workers** who value **self-reliance and freedom of movement** but experience **security systems that assume a home**, NOMAD.OS helps them **arrive in any country already set up, and recover from anywhere** by **downloading one profile per country that sets four layers: network, money, numbers, and a safety net**.

Evidence table, 1 of 3 E1 to E9

ID	SOURCE	OBSERVATION	INTERPRETATION	VALUE	TENSION
E1	Creator setup guides, Jul and Oct 2026	Gear-by-gear stacks: a named travel router running WireGuard, multiple eSIMs, priced and weighed per layer.	The setup itself is a display of competence.	Preparedness, control	Preparedness vs. burden and cost
E2	Nomad tools blog, Sep 2026	Argues VPN value depends on the work; light, non-sensitive work on a personal hotspot may not need one.	Nuance entering a VPN-maximal conversation.	Practicality	Fear vs. actual need
E3	VPN vendor blog, Aug 2026	Opens by saying most public Wi-Fi guides sell fear or sell a VPN; calls a VPN often overkill for low-risk use.	Even sellers distance themselves from fear marketing.	Calm competence	Marketing vs. trust
E4	VPN vendor blog, two posts, Jul and Aug 2026	One post says HTTPS protects you from hackers on public Wi-Fi; the other says HTTPS alone won't keep you safe.	Guidance is unstable even within one source.	Clarity	Conflicting advice
E5	VPN blog, Jul 31, 2026; travel blog comments, 2026	Readers describe buying travel routers so an employer sees a home IP; the vendor warns detection is likely.	Security gear doubles as concealment.	Freedom	Protection vs. concealment
E6	Travel blog post and reader comment, Jul 2026	A perpetual traveler's phone is stolen; a reader notes phones are grabbed while unlocked in use.	Risk peaks at the moment of use, not at rest.	Vigilance	Convenience vs. safety
E7	Phone accessory brand blog, Aug 2026	Recommends carrying a secondary "burner" phone for risky situations abroad.	Redundancy extends to whole devices.	Preparedness	More gear vs. simplicity
E8	Hacker News, Ask HN, Sep 2026 (not travel-specific)	After a phone theft, the poster can't reach banking, email, or cloud files. Replies: keep two signed-in devices; keep codes in a password manager.	One device is a single point of failure; peers prescribe redundancy.	Self-reliance, peer help	Security vs. access
E9	Q&A help site, Oct 2026 (traveler, not confirmed remote worker)	Traveler in New Zealand is locked out of Gmail; her recovery number is an Australian line she can't use.	Recovery paths assume you're near home.	Access	Mobility vs. verification

Evidence table, 2 of 3 E10 to E18

ID	SOURCE	OBSERVATION	INTERPRETATION	VALUE	TENSION
E10	Google Account Help, current page (context)	Recovery tips: use a familiar device and a location where you usually sign in, like home or work.	Platform policy encodes a home.	n/a	Home assumption
E11	Apple Support, current page (context)	Stolen Device Protection adds biometric checks and a delay when the phone is away from familiar locations.	The same assumption, in theft protection.	n/a	Home assumption
E12	r/digitalnomad thread, Apr 22, 2026 (also summarized in May 2026)	About $68k held by a fintech app for nearly 6 months; 230+ comments. Replies: it isn't a bank, keep a real one; regulator complaints released funds. Update: money returned late May.	Failure stories become community rules.	Collective knowledge	Convenience vs. control of money
E13	Expat VPN site, Oct 2026; nomad tax guide, Jun 2026	One says a dedicated-IP VPN ends repeated selfie checks; the other says VPNs can trigger fraud flags and freezes.	The same tool is sold as cure and blamed as cause.	Access	Conflicting advice
E14	Expat phone setup guide, May 2026	Banks and government portals send codes only to real US numbers, mostly not VoIP; recommends a cheap carrier line kept active from abroad.	Nomads rebuild a "home" number.	Continuity	Freedom vs. home dependence
E15	Nomad blog, Apr 2026 (8 years nomadic)	Calls the password manager the keystone; keeps a printed emergency kit sealed with a trusted person at home; warns against carrying the only copy.	Recovery depends on people and place.	Care, preparedness	Freedom vs. home dependence
E16	r/digitalnomad, ~Jun 2026, 70+ comments	PSA that the phone is the key to everything; replies advise storing IMEI and carrier details in a password manager; one reply also locked out of Google.	The phone is a single point of failure, and the community knows it.	Preparedness	Security vs. access
E17	r/digitalnomad, ~Aug 2026, 40+ comments	Thread on carrying a secondary phone; the top answer adopted it after a phone snatch in Barcelona.	Redundancy is learned through failure.	Preparedness	More gear vs. simplicity
E18	r/digitalnomad, May to Oct 2026, three threads	Full-time travel "ruined" a poster's number; replies split life across two numbers; the keep-your-US-number question recurs and draws weary replies.	The home number is infrastructure nomads must maintain from afar.	Continuity	Freedom vs. home dependence

Evidence table, 3 of 3 E19 to E27

ID	SOURCE	OBSERVATION	INTERPRETATION	VALUE	TENSION
E19	r/digitalnomad, ~May 2026, 160+ comments	A newcomer about to quit and travel is told to carry two phones and absolutely keep a home number for bank and government codes.	Status is passed down as rules for newcomers.	Belonging, care	Freedom vs. home dependence
E20	r/digitalnomad, Apr to Aug 2026, 120 to 390+ comments	Stealth-nomading threads reference the subreddit wiki setup (travel router, tunnel home, kill switch); a reply says IT reads a foreign login as stolen credentials.	Concealment is organized and documented, not fringe; corporate security also assumes a home.	Freedom, control	Protection vs. concealment
E21	r/digitalnomad, ~May 2026, 10 to 20+ comments	Newcomers ask whether a $10k transfer will trigger a freeze, and report transfers held for extra security checks.	Failure stories shape behavior before failure happens.	Control of money	Convenience vs. control
E22	X, Sep 21, 2026, creator (~23.5k followers), 367 likes	Phone stolen by motorbike thieves in Asunción; cards added to Apple Pay and thousands taken; 2FA tied to the stolen phone and a Spanish SIM, so he flew to Spain to recover the number.	Recovery can require physically returning to a “home” country.	Self-reliance	Freedom vs. home dependence
E23	X, Oct 5 to 8, 2026, three small accounts	One says get your cell number out of 2FA and use Google Voice; one says banks may refuse Voice numbers, so test each bank; one says keep the home SIM on just for codes.	Advice conflicts in the same week, with no shared answer.	Clarity	Conflicting advice
E24	X, Oct 6, 2026, developer creator (~400k)	Praises a GL.iNet travel router for making travel easier; a VPN is optional.	The router is also a mainstream convenience gadget, not only a security or concealment tool.	Convenience	(Complicates the burden tension)
E25	X, Oct 1, 2026, small nomad account	Describes briefing an eSIM provider on 2FA, eSIM, and the financial risks of a stolen phone.	The market is starting to notice the same gap.	n/a	Competitive signal
E26	r/digitalnomad, ~May 27, 2026, 38 upvotes, 39 comments	A work account was frozen after using a consumer VPN while traveling; replies say commercial VPNs are easy to detect and suggest self-hosting at a parent’s house.	The security tool triggers the lockout, and the fix routes back through home.	Access	Security vs. access
E27	r/GlInet, ~Apr 8, 2026, 13 comments (verified via a listing)	A newcomer offers to pay for help setting up a travel router to present a home IP; replies cover home-side servers, 2FA, and disabling Wi-Fi on work laptops.	Concealment setups are hard enough that newcomers pay for help.	Freedom	Protection vs. concealment

APPENDIX D · FROM ASSIGNMENT 4

Research-to-design traceability.

EVIDENCE	WHAT WE OBSERVED OR CONCLUDED	DESIGN RESPONSE
E10 E11 E22	Recovery keys off home; one nomad flew home to get a number back.	Safety net: Break Glass from any browser; a Keyholder replaces the familiar-place check. (S6, S7)
E13 E23 E26	The right answer depends on the country and the bank.	Country profiles, with dated reports for each country. (S1, S2)
E2 E3 E4	VPNs: needed versus overkill, and fear-driven advice.	Network layer: VPN on only for untrusted Wi-Fi. (S3)
E13 E26	VPNs trigger bank and work-account freezes.	Network layer: apps that flag VPNs go direct. (S3)
E6 E22	Thieves add cards to wallets and move money fast.	Money layer: cards locked to trip dates and places. (S4)
E12 E21	Fintech freezes strand money for months.	Money layer: a backup way to pay and a freeze plan. (S4)
E14 E18 E19	Nomads keep a home number alive for codes.	Numbers layer: home line for codes, local eSIM for data. (S5)
E8 E23	Advice splits on SMS versus other code methods.	Numbers layer: codes that can leave SMS move to an authenticator. (S5)
E15 E17	Nomads leave kits with family and add backups after a theft.	Safety net: a Keyholder with a split key. (S6)
E5 E20	Security gear is used to hide location from employers.	Principle: encrypt, don’t disguise. The VPN exits where you are. (Principles)

Sources for every evidence item.

E1	dev.to/notifyshow/the-ultimate-digital-nomad-setup-minimalist-tech-and-working-from-anywhere-179k; nomadstoolslab.com/digital-nomad-setup-guide-2026
E2	ontheroadtools.com/vpn-for-digital-nomads-do-you-actually-need-one
E3	windscribe.com/blog/how-to-use-public-wi-fi-safely
E4	protonvpn.com/blog/vpn-myths; protonvpn.com/blog/public-wifi-and-https
E5	vpn.pro/blog/can-your-employer-detect-that-youre-using-a-vpn-while-working-remotely; indietraveller.co/best-travel-vpn (comments)
E6	johnnyjet.com/dont-fall-for-this-new-phone-theft-scam-targeting-tourists
E7	phoneloops.com/blogs/posts/phone-security-while-traveling-europe-prevent-theft-stay-safe-in-2026
E8	news.ycombinator.com/item?id=49742976
E9	justanswer.com/email/jeery-locked-gmail-account-currently-overseas-none.html
E10	support.google.com/accounts/answer/7299973
E11	support.apple.com/120340
E12	reddit.com/r/digitalnomad/comments/1ssjy5m; summary at moveandmoney.com/blog-fintech-account-frozen-2026
E13	safeexpatvpn.com/revolut-account-restricted-abroad; nomadtaxguide.com/blog/digital-nomad-banking-guide
E14	expatretireguide.com/money/phone-setup
E15	tonydehnke.com/blog/digital-nomad-backup-tools
E16	reddit.com/r/digitalnomad/comments/1uh3o4f
E17	reddit.com/r/digitalnomad/comments/1vwj8pg
E18	reddit.com/r/digitalnomad/comments/1w75mkf; 1wyr9jc; 1t8q1ig
E19	reddit.com/r/digitalnomad/comments/1ti45hs
E20	reddit.com/r/digitalnomad/comments/1vzmfww; 1tr5hou; 1slwlzs
E21	reddit.com/r/digitalnomad/comments/1tqfsq4; 1tlf6pc
E22	x.com/hispanicnomad/status/2102094776187134436
E23	x.com/GabeLenz/status/2108298536093896991; x.com/StephenZhou1004/status/2108115835294216253; x.com/nexkit2026/status/2107078468936908918
E24	x.com/theo/status/2107590392128704776
E25	x.com/baddigitalnomad/status/2105673102768419026
E26	reddit.com/r/digitalnomad/comments/1tp3a7b
E27	reddit.com/r/GllNet/comments/1sfksl9

Reddit items come from Google listings filtered to the study window, dates estimated from post IDs; X posts were opened and checked by hand. Dates are publication or comment dates where shown; otherwise estimated from the page's indexed age and given to the month. Policy pages were read in October 2026. Usernames are omitted throughout.

APPENDIX F

References.

Airalo. (2026, June 2). *Airalo surpasses 30 million users* [Press release]. https://www.airalo.com/about-us/press-releases/airalo-surpasses-30-million-users	Existing product analysis
Amershi, S., Weld, D., Vorvoreanu, M., Fourney, A., Nushi, B., Collisson, P., Suh, J., Iqbal, S., Bennett, P. N., Inkpen, K., Teevan, J., Kikin-Gil, R., & Horvitz, E. (2019). Guidelines for human-AI interaction. *Proceedings of CHI 2019*. https://doi.org/10.1145/3290605.3300233	Explain, confirm, and support correction
Google PAIR. (2021). *People + AI guidebook*. https://pair.withgoogle.com/guidebook	Trust, feedback, and errors
OpenAI. (2025, January 23). *Introducing Operator*. https://openai.com/blog/introducing-operator	Precedent: an agent that asks before it buys
Parasuraman, R., Sheridan, T. B., & Wickens, C. D. (2000). A model for types and levels of human interaction with automation. *IEEE Transactions on Systems, Man, and Cybernetics, Part A, 30*(3), 286-297. https://doi.org/10.1109/3468.844354	How much the agent does on its own
Shneiderman, B. (2022). *Human-centered AI*. Oxford University Press.	High automation with high human control
Federal Trade Commission. (n.d.). *Avoid scams when you travel*. Consumer Advice. https://consumer.ftc.gov/articles/avoid-scams-when-you-travel	Wire transfers as a scam signal
NOMAD.OS team. (2026). *Assignment 4 research: 27 public posts on how nomads stay secure, E1 to E27*. IDSN 523.	Evidence codes on slide 3, in full in Appendix C